

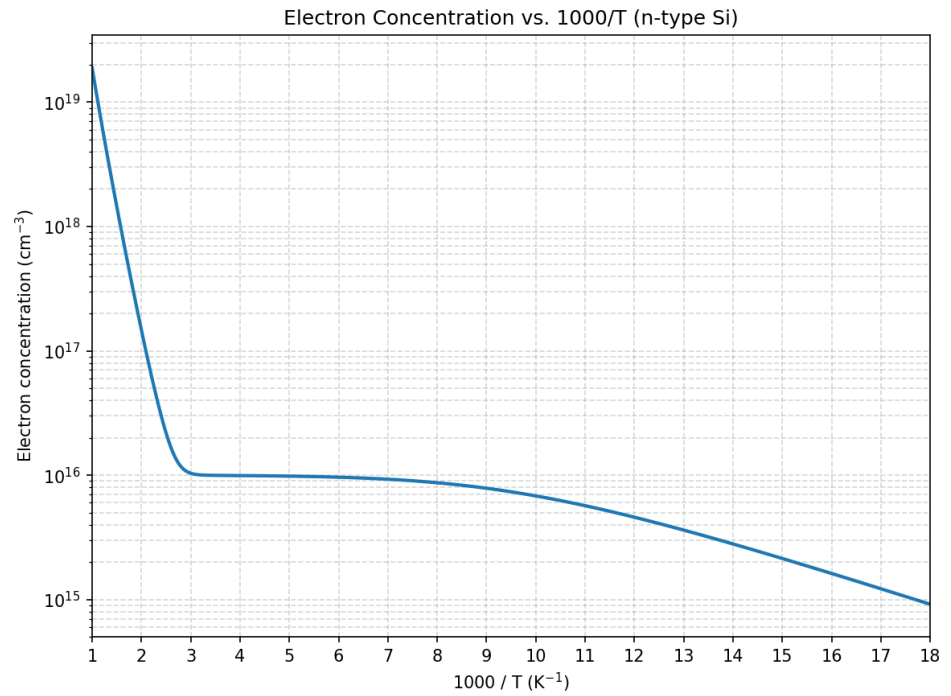
Semiconductor Device Fundamentals Exam 1

INSTRUCTION: Do your work neatly and *write key steps only*. If you are making an assumption please state it clearly and justify it. Don't make any unjustified approximations. You may want to work out the solution in rough sheet and only then write it in the space provided below

Constants

Parameters	Values	Units	Parameters	Values	Units
ϵ_0	8.85×10^{12}	F/m	$\hbar$	1.05×10^{-34}	J-sec
ϵ_{Si}	11.7	ϵ_0	k_B	1.38×10^{-23}	J/K
ϵ_{SiO_2}	3.9	ϵ_0	n_i	1.5×10^{16}	m^{-3}
q	1.6×10^{-19}	C	T	300	K

- Consider a semiconductor with doping N_D . The temperature of the semiconductor is swept from low to high and the resulting carrier concentration is shown below. Extract approximate values of (1) the donor energy level with respect to the conduction band edge (2) the doping concentration, and (3) the bandgap of the material. Assume that bandgap is temperature independent [Marks: 2+1+2]



2. Consider an n -type conductor bar with non-uniform doping ($N_D = N_0 e^{-x/L}$) in thermal equilibrium. What will be the value of the electric field inside the semi-conductor. Neglect the incomplete ionization, recombination and generation. Provide briefly physical picture of the situation inside the semiconductor bar. [**Marks: 3+2**]

3. In an n -type semiconductor bar of length L and $N_D = 10^{18} \text{ cm}^{-3}$, we steadily inject only excess holes at $x=0$ having a concentration of $5 \times 10^{16} \text{ cm}^{-3}$. Assuming hole mobility to be $500 \text{ cm}^2/\text{V} - \text{s}$ and electron mobility to be $1400 \text{ cm}^2/\text{V} - \text{s}$ with the excess carrier lifetime being $1\mu\text{s}$. Calculate (1) the separation in the hole quasi Fermi-level and valence band at $x = 100 \text{ nm}$ (assuming $L \gg 100\text{nm}$). (2) Hole current at $x = 0$ and $x = 100 \text{ nm}$. Given: $m_{n,DOS}^* = m_{p,DOS}^*$, $E_G = 1.1\text{eV}$ (**Marks 3 + 2**)

4. Consider a sample of an n-type silicon with $N_D = 10^{15} \text{ cm}^{-3}$. The sample is uniformly excited such that the generation rate is $10^{19} \text{ cm}^{-3} \text{ sec}^{-1}$ and has reached a steady state. Assume that the excess carrier life-time is $10 \mu\text{s}$. Calculate (1) the separation in the electron and hole quasi-Fermi levels in eV and (b) change in the conductivity of the sample due to the shining of light. Consider $\mu_n = 1300 \text{ cm}^2/\text{V} - \text{s}$ and $\mu_p = 463 \text{ cm}^2/\text{V} - \text{s}$ (**Marks: 3 + 2**)